Where $\lambda = np$

- The mean and variance of the poisson distribtution is λ.
- The λ is the only parameter of poisson distribution.
- Poisson distribution can never be symmetrical.
- It is a distribution for rare events.
- Normal distribution is the limiting form of binomial distribution when n is large and neither p nor q is small
- The normal probability distribution is given by $f(x)\ \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}$
- The mean of the distribution is μ
- The SD of the distribution is σ.
- It is a symmetrical distribution
- The graph of the distribution is bell shaped
- In normal distribution the mean, median and mode are equal
- The points of inflexion are $\mu - \sigma$ and $\mu + \sigma$
- The normal curve approaches the horizontal axis asymptotically
- Area Property : In a normal distribution about 68% of the item will lie between $\mu - \sigma$ and $\mu + \sigma$. About 95% will lie between are $\mu - 2\sigma$ and $\mu + 2\sigma$. About 99% will lie between $\mu - 3\sigma$ and $\mu + 3\sigma$.
- Standard normal random variate is denoted as $Z = (X - \mu)/\sigma$
- The standard normal probability distribution is $1/\sqrt{2\pi}(e^{\frac{-z^2}{2}})$
- The mean of the distribution is zero and *SD* is unity
- The points of inflexion are at $z = -1$ and $z = +1$

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